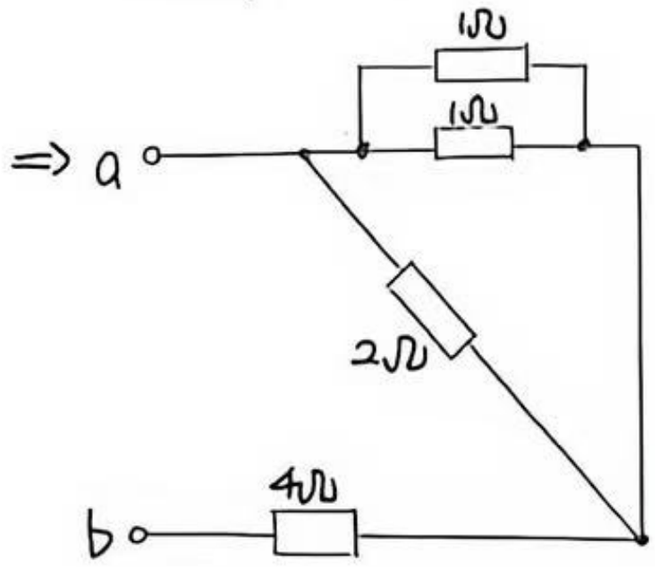
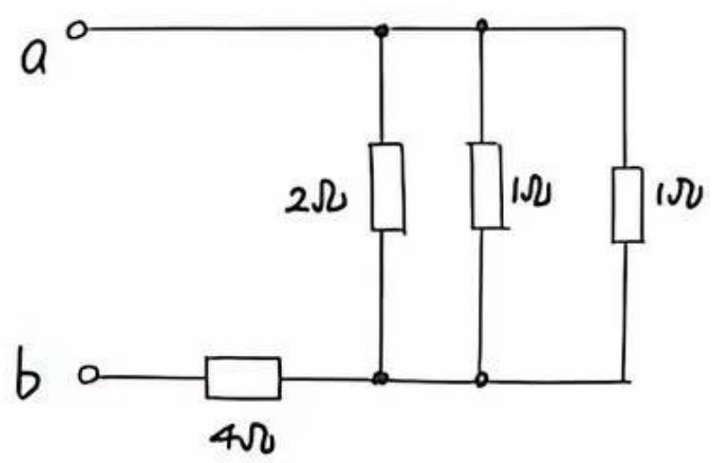
2025级电路解答与评注

一、填空

1. 显然最右侧1Ω电阻被短路：



$\Rightarrow$



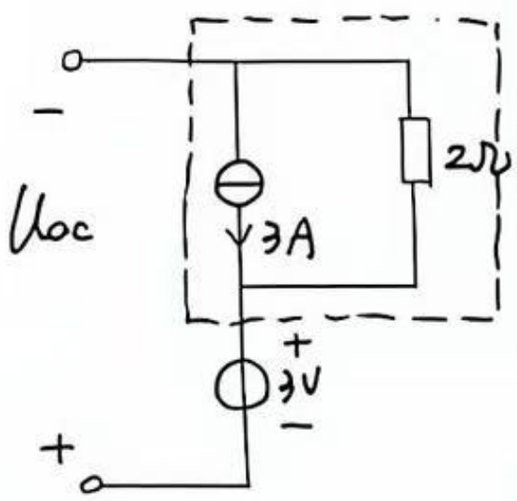
$$\therefore R_{ab} = 4 + \frac{1}{\frac{1}{2}+\frac{1}{1}+\frac{1}{1}}$$

$$= 4 + \frac{2}{5}$$

$$= \boxed{4.4\Omega}$$

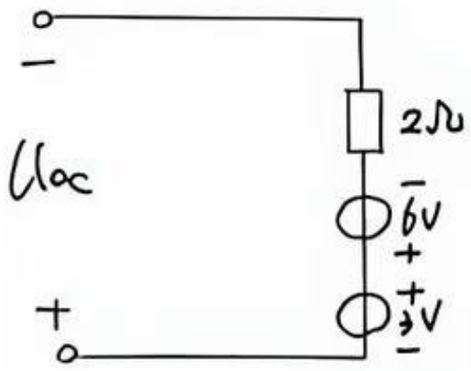
2. 右下角的3A电流源与3V电压源并联，对外失效：

$\Rightarrow$

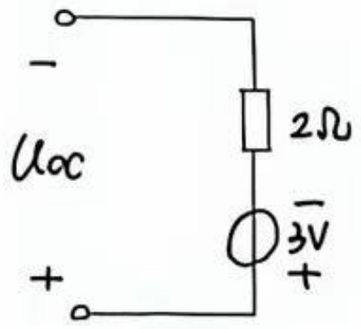


$\Rightarrow$ 有伴电流源与有伴电压源之间相互等效：

$\Rightarrow$



$\Rightarrow$



$$\boxed{U_{oc} = 3V \quad R_{eq} = 2\Omega}$$

3. 全响应的两种分解：$f(t) = \underbrace{f(\infty)}_{\text{稳定量}} + \underbrace{[f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}}_{\text{暂态量}}$

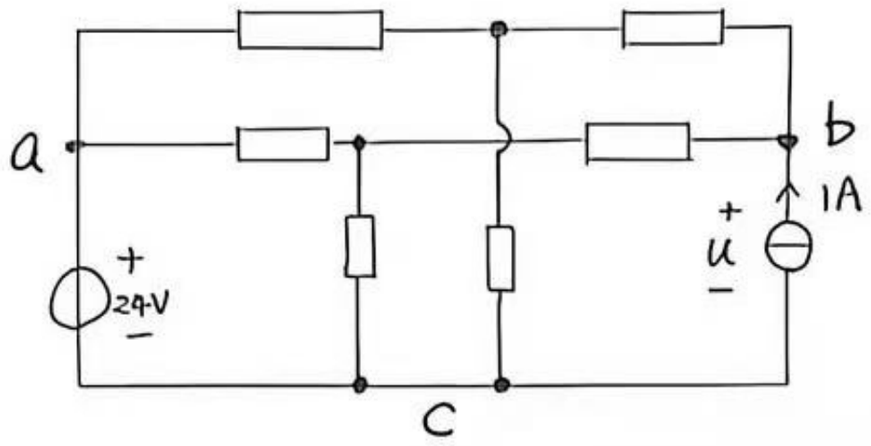
$$= \underbrace{f(0^+)e^{-\frac{t}{\tau}}}_{\text{零输入}} + \underbrace{f(\infty) - f(\infty)e^{-\frac{t}{\tau}}}_{\text{零状态}}$$

列方程：$f(\infty) = 6$

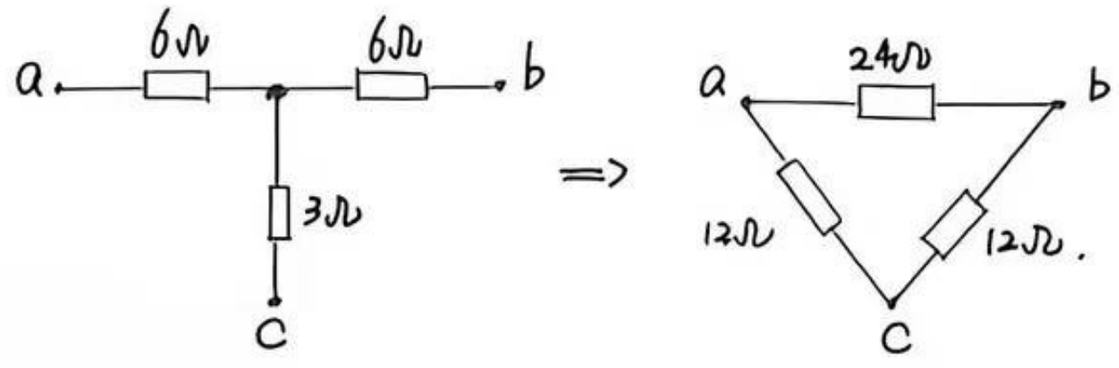
$f(0^+) - f(\infty) = 8$

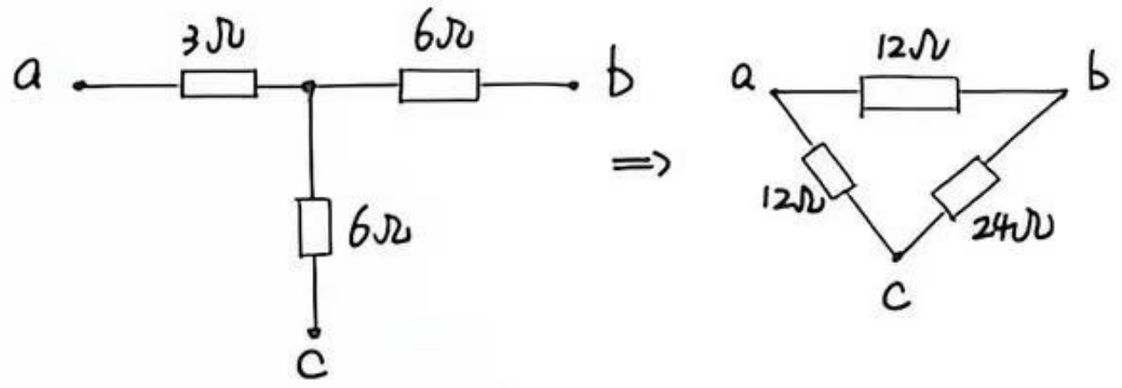
$\because f(0^+) = 14 \quad \therefore$ 零输入：$\boxed{14e^{-5t}V \quad (t \geq 0)}$

4.



先考虑用"Y-Δ"变换改善电路拓扑结构:



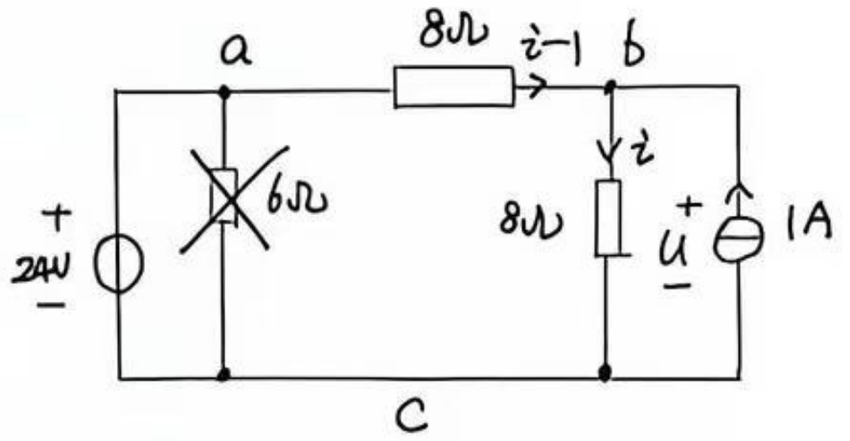


∃

$\Rightarrow R_{ab} = 24\Omega // 12\Omega = 8\Omega.$ $\Rightarrow$

$R_{ac} = 12\Omega // 12\Omega = 6\Omega.$

$R_{bc} = 12\Omega // 24\Omega = 8\Omega.$

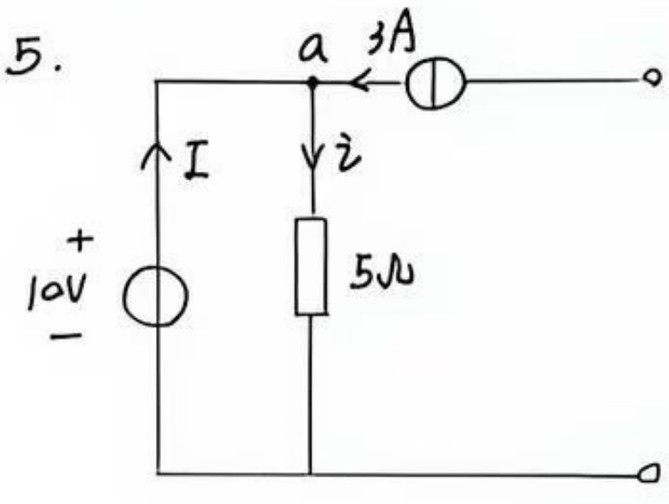


6Ω电阻与24V电压源并联. 失效

设右侧8Ω电阻支路电流为i. 对b点列KCL, 上方8Ω流过电流为i-1,

对左侧网孔列KVL: $24 = 8(i-1) + 8i \quad \therefore i = 2A \quad \therefore U = 8i = \boxed{16V}$

5.



5Ω电阻流过电流 $i = \frac{10V}{5\Omega} = 2A$

对a点列KCL: $I + 3 = 2$

$\therefore I = \boxed{-1A}$

$P_{源} = 10 \times (-1) = -10W$, 非关联参考方向, 吸收 $\boxed{10W}$

6. 思路: 写出端口的等效阻抗, 然后令虚部为0

$0.5H \rightarrow j\cdot\frac{1}{2}w$; $1\mu F \rightarrow \frac{1}{j\cdot 10^6\cdot w} = \frac{10^6}{jw}$.

$\therefore Z_{ab} = j\cdot\frac{1}{2}w + \frac{10^3\cdot\frac{10^6}{jw}}{10^3+\frac{10^6}{jw}}$, 虚部为: $\frac{w}{2} + \frac{-w\cdot 10^6}{w^2+10^6} = 0$

$\Rightarrow \boxed{w_0 = 1000\ rad/s}$

7. 顺接：$L_{顺} = L_1 + L_2 + 2M = 2.5H$ ； $Z_{顺} = j\cdot 2.5\cdot\omega$

反接：$L_{反} = L_1 + L_2 - 2M = 0.5H$ ； $Z_{反} = j\cdot 0.5\cdot\omega$

$\dot{I}_{反} = \frac{\dot{U}}{Z_{反}}$ ， $\dot{I}_{顺} = \frac{\dot{U}}{Z_{顺}}$ 显然 $\dot{I}_{反} : \dot{I}_{顺} = \boxed{5:1}$

8. 由于三角函数的正交性，平均功率 $P = U_0 I_0 + \sum_{i=1}^{\infty} U_K I_K \cos\varphi_K$ $(\varphi_K = \varphi_{uK} - \varphi_{iK})$

直流：$P_0 = 10\times 3 = 30W$

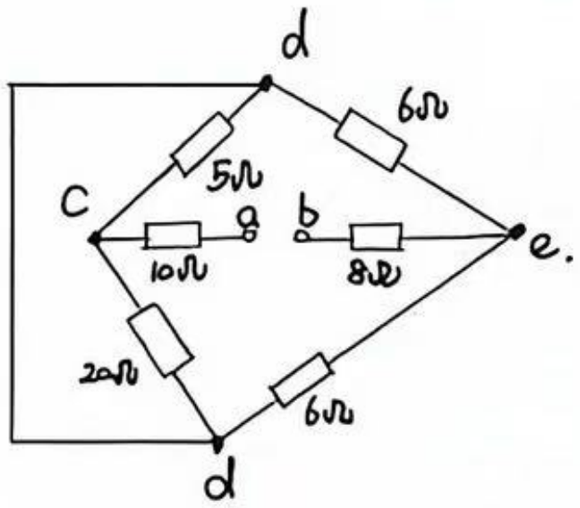
基波：$P_1 = \frac{20}{\sqrt{2}}\cdot\frac{6}{\sqrt{2}}\cdot\cos(-30°-30°) = 30W$ $\therefore P = P_0 + P_1 + P_3 + P_5$

三次谐波：$P_3 = \frac{30}{\sqrt{2}}\cdot 0 = 0$ $= \boxed{60W}$

五次谐波：$P_5 = \frac{10}{\sqrt{2}}\cdot 0 = 0$

二.

图(a)：



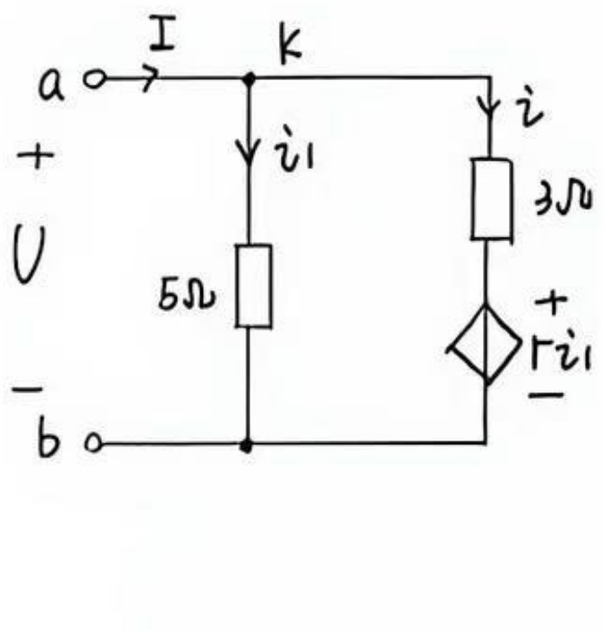
标等电位点、c，d，e：

$R_{cd} = 5\Omega // 20\Omega = 4\Omega$

$R_{de} = 6\Omega // 6\Omega = 3\Omega$.

$\therefore R_{ab} = 10\Omega + 4\Omega + 3\Omega + 8\Omega = \boxed{25\Omega}$

图(b)：



加压求流法：

$U = 5i_1$

对右侧网孔列 KVL：

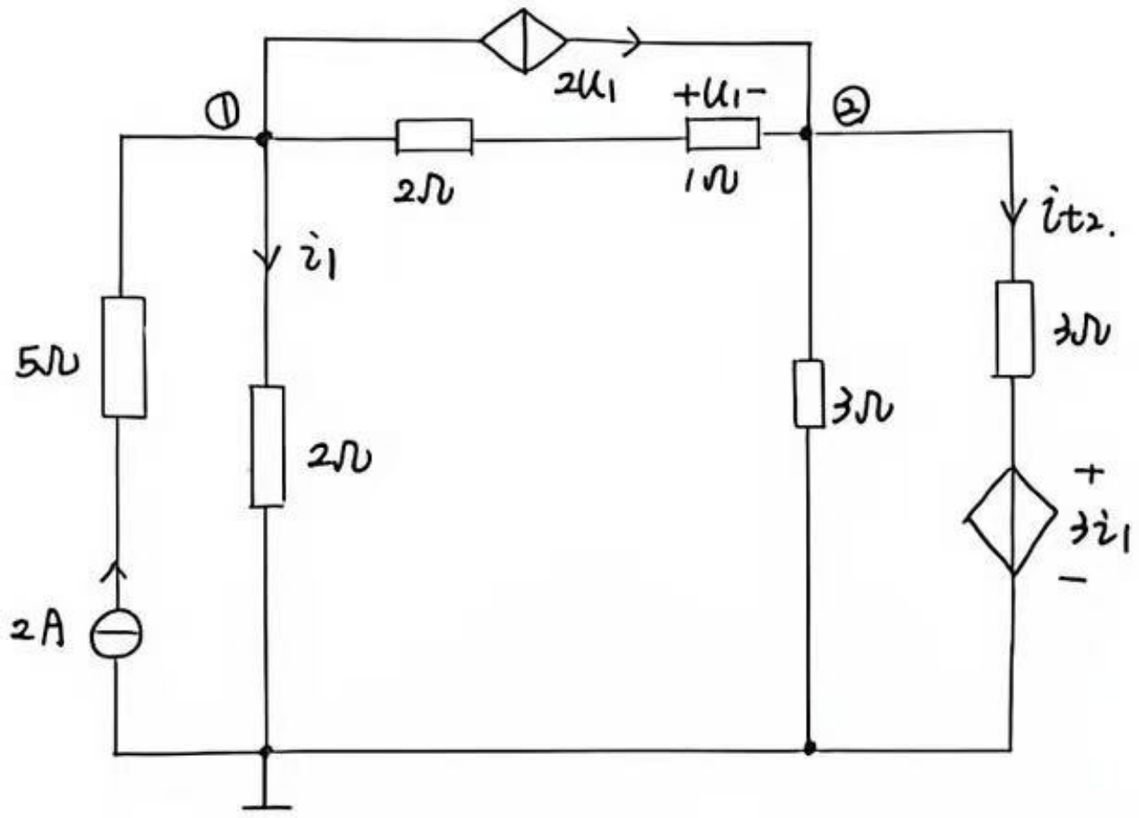
$5i_1 = 3i + ri_1$ $\therefore$ $i = \frac{(5-r)i_1}{3}$

对节点 K 列 KCL：

$I = i_1 + i = i_1 + \frac{(5-r)i_1}{3} = \frac{(8-r)i_1}{3}$

$R_{ab} = \frac{U}{I} = \boxed{\frac{15}{8-r}}\ (\Omega)$.

三、 最左侧5Ω电阻与2A电流源串联，对外失效（但对内不失效！）



节点①：

$$u_{n1}\left(\frac{1}{2}+\frac{1}{3}\right)-u_{n2}\cdot\frac{1}{3}=2-2u_1 \qquad (1)$$

节点②：

$$u_{n2}\left(\frac{1}{3}+\frac{1}{3}+\frac{1}{3}\right)-u_{n1}\cdot\frac{1}{3}=2u_1+i_1 \qquad (2)$$

增补方程：

$$i_1=\frac{u_{n1}}{2} \qquad (3)$$

$$u_1=\frac{u_{n1}-u_{n2}}{3} \qquad (4)$$

解(1)(2)(3)(4)得：

$$\begin{cases} u_{n1}=\frac{10}{3}V \\ u_{n2}=3V \\ i_1=\frac{5}{3}A \\ u_1=\frac{1}{9}V \end{cases}$$

上面的受控源：

$$u_{t1}=u_{n1}-u_{n2}=\frac{1}{3}V$$

$$I_{t1}=2u_1=\frac{2}{9}A$$

$$P_{t1}=\frac{1}{3}\times\frac{2}{9}=\boxed{\frac{2}{27}W\ (\text{吸收})}$$

下面的受控源：

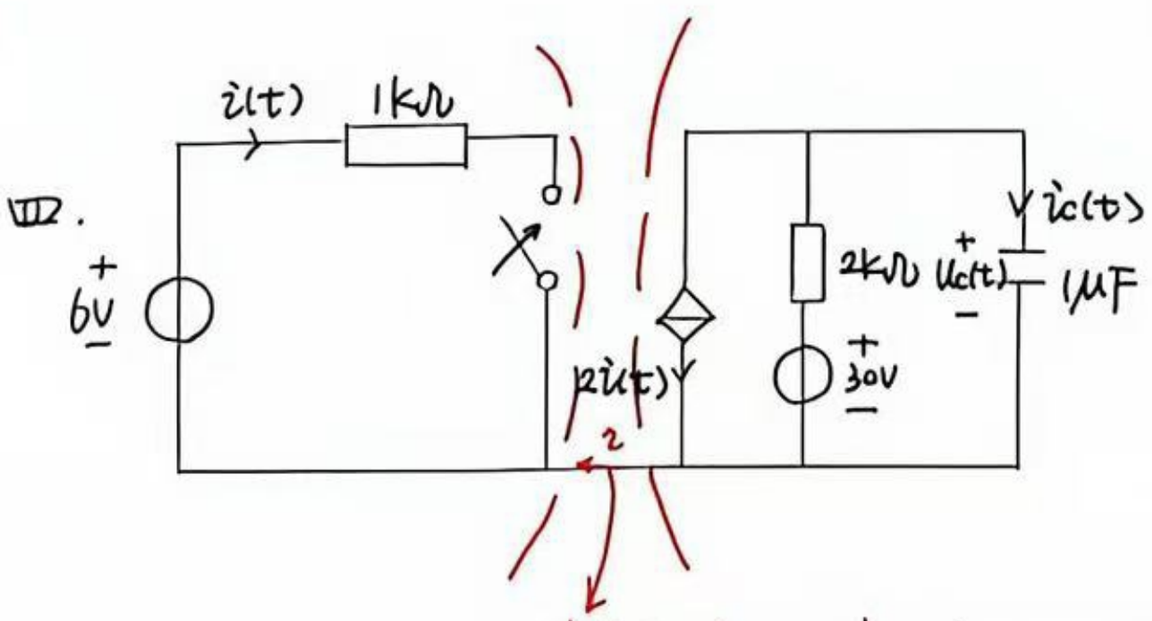
$$u_{t2}=3i_1=5V$$

对最右侧网孔列KVL：

$$3=3i_{t2}+5 \quad \therefore\ i_{t2}=-\frac{2}{3}A$$

$$P_{t2}=5\times\left(-\frac{2}{3}\right)=\boxed{-\frac{10}{3}W\ (\text{吸收})}$$

四.



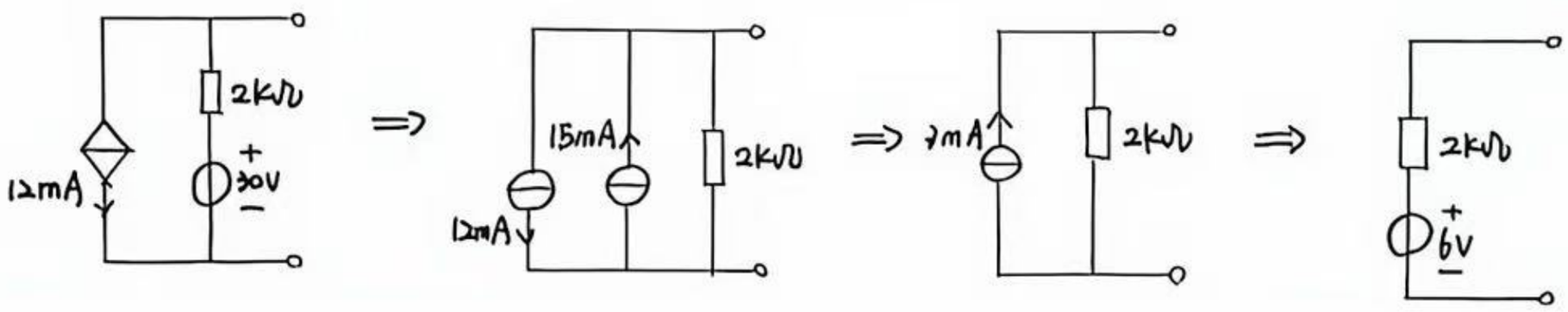
闭合前，2kΩ电阻失效，$u_C(0^-)=30V$

闭合后，由换路定理，$u_C(0^+)=u_C(0^-)=30V$

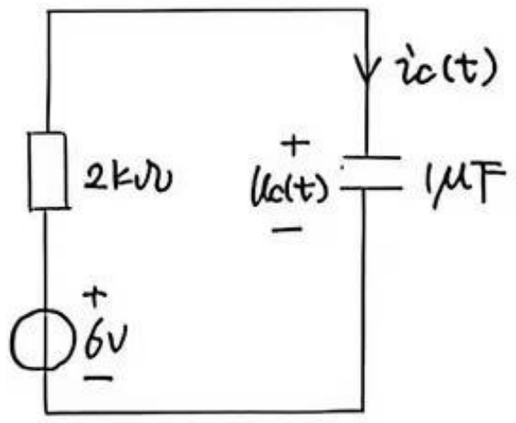
且左，右两部分独立，互不影响

该单导线上电流一定为0（可以对左，右任一个节点列KCL，不难发现$\sum i=i=0$）

$\Rightarrow i(t)=\frac{6V}{1k\Omega}=6mA$，对右侧嵌套结构戴维南：



⇒.



$$\tau = R_{eq} \cdot C = 2\cdot10^{3}\cdot10^{-6} = 2\cdot10^{-3}\ s$$

$$u_C(\infty) = 6V$$

三要素法：

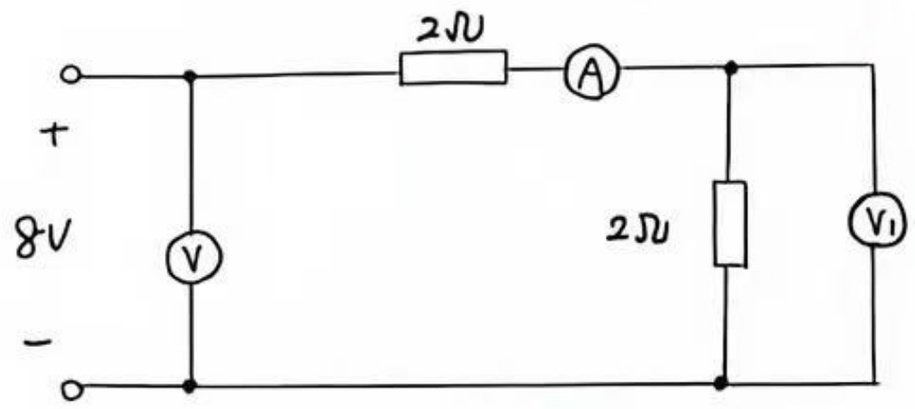
$$u_C(t) = u_C(\infty) + [u_C(0^+) - u_C(\infty)]\,e^{-\frac{t}{\tau}}$$

$$= 6 + 24e^{-500t}\ V$$

$$i_C(t) = C\cdot\frac{du_C(t)}{dt} = 10^{-6}\cdot 24\cdot(-500)e^{-500t}\ A$$

$$= \boxed{-0.012e^{-500t}\ A}$$

五. 直流情况下，LC串联当断路：

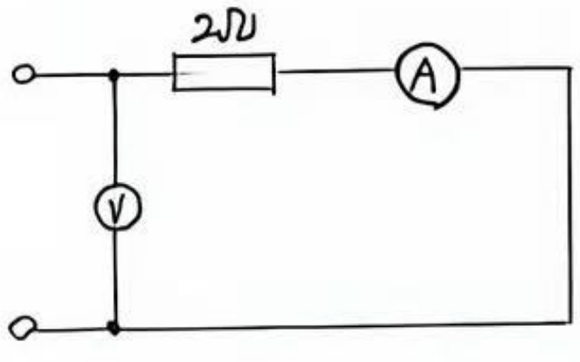


电表读数：Ⓐ：$I_1 = \frac{8}{2+2} = 2A$

Ⓥ：$U_1 = 8V$

Ⓥ1：$U_{11} = 4V$

3W谐波情况下：由于 $j\cdot3\omega\cdot L = j3\Omega$；$\frac{1}{j\cdot3\omega C} = -j3\Omega$，发生串联谐振，对外等效为短路.



电表读数：Ⓐ：$I_2 = \sqrt{I^2 - I_1^2} = 2A$

Ⓥ：$U_2 = 2I_2 = 4V$

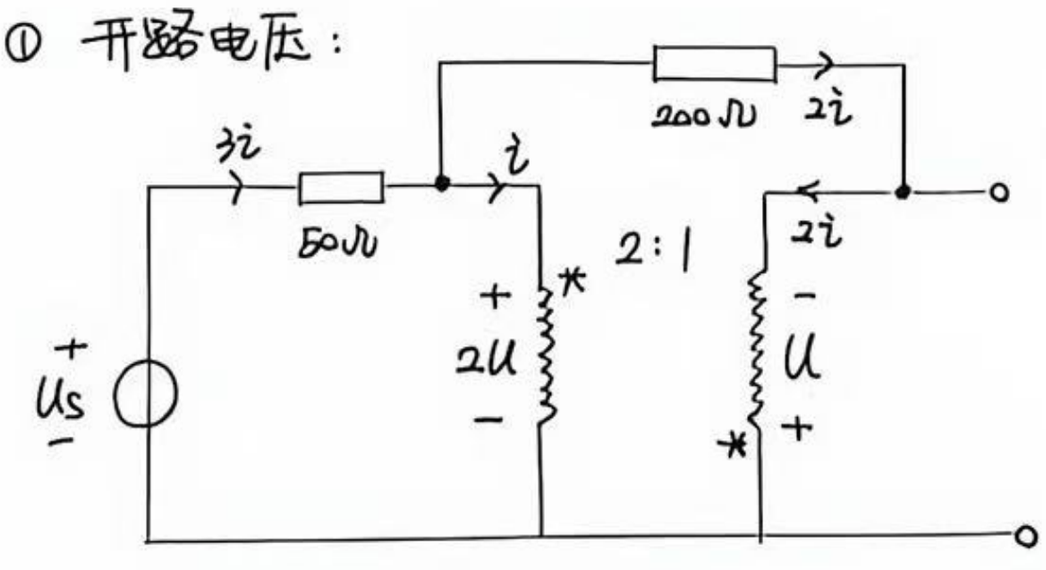
Ⓥ1：$U_{12} = 0V$

⇒ Ⓥ1：$\sqrt{4^2+0^2} = \boxed{4V}$

Ⓥ：$\sqrt{8^2+4^2} = \boxed{4\sqrt{5}V}$

六："理想变压器 + 闭合回路"估构：用"开路电压 + 短路电流"戴维南：

① 开路电压：



标出电流与电压：

左侧网孔KVL：

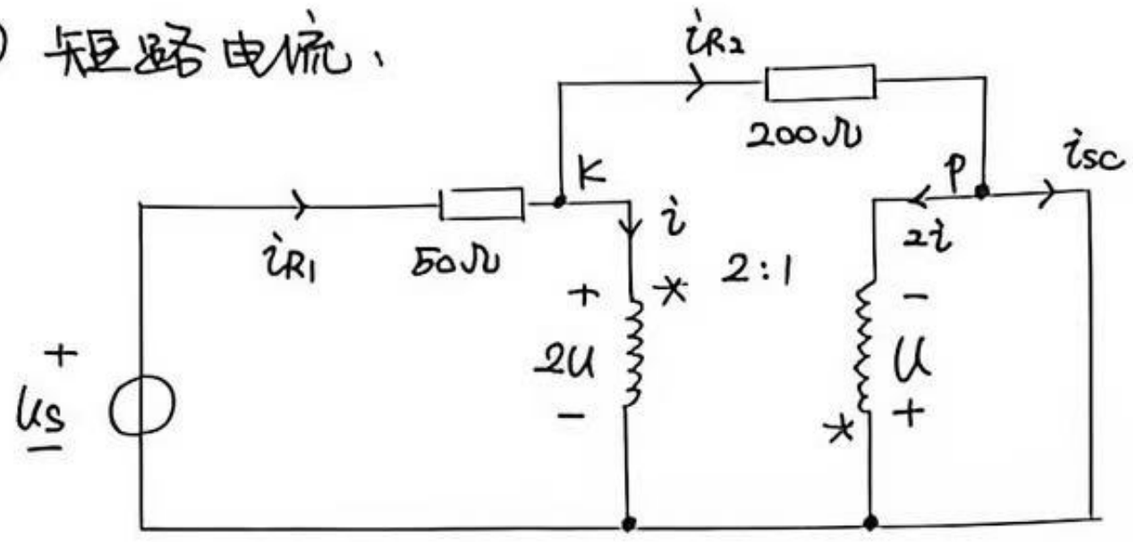
$u_s = 150i + 2u$.

右网孔KVL：

$2u + u = 400i$

$\therefore u = \frac{8}{3}u_s = \frac{352\sqrt{2}}{5}\sin 314t.\ V$

② 短路电流.



对左网孔 KVL：

$U_S = 50 i_{R_1} + 2U$, (1)

对中间网孔 KVL：

$3U = 200 i_{R_2}$ (2)

对 K 点，列 KCL

$i_{R_1} = i_{R_2} + i$ (3)

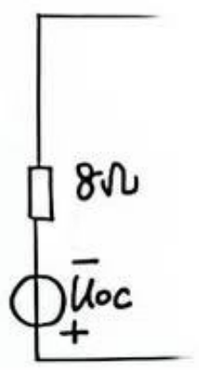
对右网孔 KVL：

$U = 0$ (4)

$\Rightarrow i = \frac{1}{50} U_S$.

对节点 P 列 KCL：$i_{sc} = -2i = -\frac{1}{25} U_S$.

$\Rightarrow$. $R_{eq} = \frac{U}{|i_{sc}|} = 8\Omega$ $\Rightarrow$ 戴维南电路：



其中 $U_{oc} = \frac{352\sqrt{2}}{5} \sin 314t$ V

$R_L = 8\Omega$ 时功率最大. $P_{max} = \frac{U_{oc}^2}{4R_{eq}} = \frac{\left(\frac{352}{5}\right)^2}{32} = \boxed{\frac{3872}{25} W}$

七. 设 N 的传输矩阵为 $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$. 由于 N 是对称线性电阻网络.

有：$A = D$, $AD - BC = 1$

当 $R_2 = \frac{1}{3}\Omega$ 时. $U_2 = 0.5V$ $\therefore$ $I_2 = \frac{U_2}{R_2} = 1.5A$

$U_1 = 2V$ $I_1 = 12 - \frac{U_1}{R_1} = 6A$

$\because \begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} U_2 \\ I_2 \end{bmatrix} \Rightarrow \begin{cases} \frac{1}{2}A + \frac{3}{2}B = 2 \\ \frac{1}{2}C + \frac{3}{2}D = 6 \end{cases} \Rightarrow \begin{cases} A = \frac{17}{8} \\ B = \frac{5}{8} \\ C = \frac{45}{8} \\ D = \frac{17}{8} \end{cases}$

$\therefore$ 传输方程为：$\boxed{\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} \frac{17}{8} & \frac{5}{8} \\ \frac{45}{8} & \frac{17}{8} \end{bmatrix} \begin{bmatrix} U_2 \\ I_2 \end{bmatrix}}$

当 $R_2=0$ 时，$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} \frac{17}{8} & \frac{5}{8} \\ \frac{45}{8} & \frac{17}{8} \end{bmatrix} \begin{bmatrix} 0 \\ I_2 \end{bmatrix}$

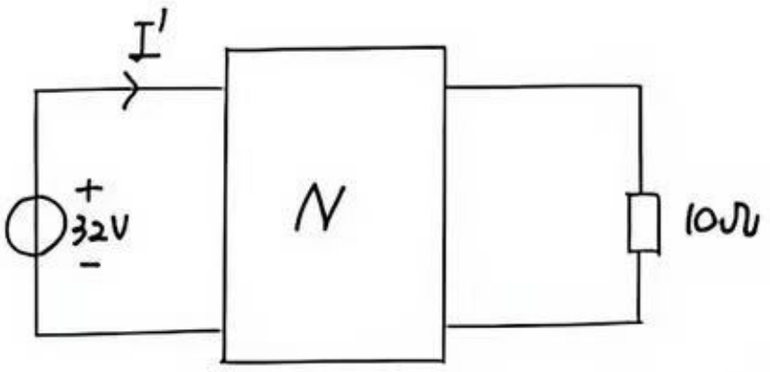
$\therefore U_1 = \frac{5}{8} I_2$ ， $I_1 = \frac{17}{8} I_2$.

左侧列 KCL：$I2 = \frac{U_1}{R_1} + I_1 = \frac{15}{8} I_2 + \frac{17}{8} I_2 = 4 I_2$ $\therefore$ $I_2 = 3A$

$U_1 = \frac{5}{8} I_2 = \frac{15}{8} V$

八. 我们应用叠加定理：

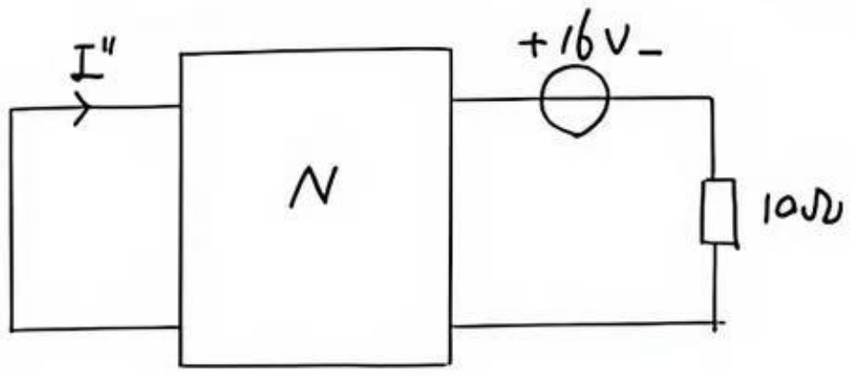
case1：



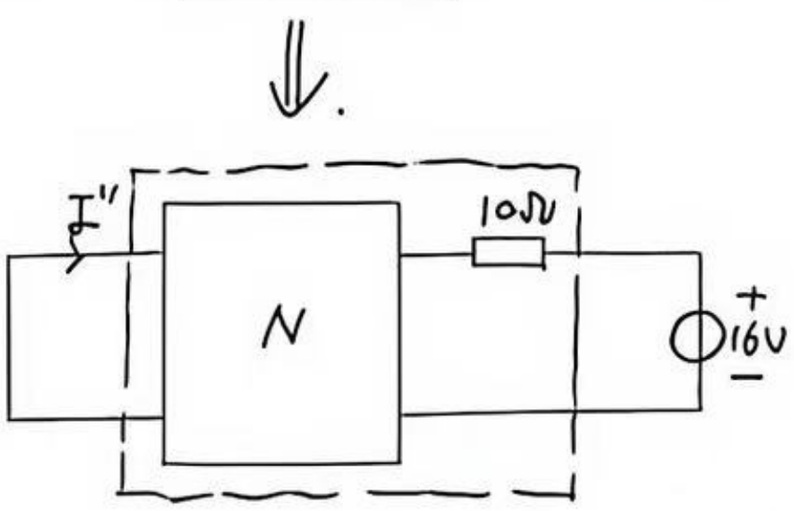
这个情况的 I' 就是图1中的 I_1.

$\therefore I' = I_1 = 6A$.

case2：



为了方便辨认结构，将电阻适当换换位置



N 是线性无源网络，所以加上一个10Ω电阻仍是一个线性无源网络 N'.

N' 满足互易定理！

$\Rightarrow \frac{32V}{I_2} = \frac{16V}{(-I'')}$ ，其中 $I_2 = 2A$.

(-I'') → 注意方向！

$\therefore I'' = -1A$.

$\therefore I = I' + I'' = \boxed{5A}$